



ARTIFICIAL INTELLIGENCE



Fig. 3: Aibo, a robotic dog with intelligent features and autonomy (Image courtesy: www.digitaltrends.com)



Fig. 4: A musician performing with robots inspired by AI (Image courtesy: www.classicalite.com)



Fig. 5: Haley Joel Osment played a robot in Steven Spielberg's movie *A.I. Artificial Intelligence* (Image courtesy: www.spielbergfanclub.com)

sequences and knowledge acquisition from genetic data.

There is no doubt that application of AI to computational molecular biology demands exceedingly interdisciplinary and complementary skills, and these are seldom provided for in most current academic curricula. Interdisciplinary is interesting, however, unless AI and computational biology communities collaborate closely, development of new methodologies and algorithms is likely to lose pace.

Gaming. Another big area in which AI plays a vital role is the gaming industry. In the 1990s, we saw the first attempts to mass-produce AI based toys and games in the form of Tamagotchi dolls, Giga Pets, first widely-released robot, Furby and much more. Later an enhanced type of domestic robot, Aibo, a robotic dog with intelligent features and autonomy, was launched.

AI has also been applied to video

games in the form of video game bots, which are designed to stand in as opponents when people are either not available or are not desired. In a game called *Left 4 Dead*, AI based director decides where enemies brood and how maps are laid out to be more or less demanding at various points of play.

It is important for gamers to feel that the characters that are inside the game are almost real. Certain games are based on neural networking technology, which is used to create characters that learn as the game progresses. Characters in a fighting game, for example, could be taught battle skills in the same manner as the humans.

Then there are computing machines available in the market, which can play master-level chess with you. These are based on some level of AI and use aggressive computation force in evaluating hundreds of thousands of different positions. To beat a world champion by sheer computing force and known reliable heuristics requires evaluation of as many as 200 million positions per second.

Music and AI. Since long, music has evolved with technology. Computer-science engineers have been trying to make computers match the activities of skilful musicians, using AI. Composition, performance, music theory and sound processing are some of the major areas on which research in music and AI is on.

Also, efforts to model music cognition with AI are generally looked upon as methods of improving our understanding of both human psychology and intellect. Once an effective model of the music listener has been accomplished, it can be improvised into a more complex model consisting of the listener, performer and composer, all put together. It then becomes a self-learning AI system. For example, a composing program can get all the required input information from its

environment by listening to musical performances. Also, this more sophisticated model can be very insightful in terms of understanding the behaviour of musicians.

Conclusion

Haley Joel Osment played a robot created with the ability to have emotions, dreams and desires in Steven Spielberg's movie *A.I. Artificial Intelligence*, a sci-fi adaptation of Pinocchio story.

There has been a certain level of resistance to AI due to the fear of the world being taken over by machines, as the gap between humans and machines becomes narrower. However, there is also the belief that a machine can never be as good as a human being in making business decisions.

There is a school of thought that believes that human-level intelligence can be achieved by writing large numbers of programs but most AI researchers believe that new and creative fundamental ideas are required. While it cannot be predicted by when human-level intelligence will be achieved, we do not really need to simulate conscious human thought as such.

The emphasis today is on developing computers that can be operated intuitively with minimum human involvement. This demands a system that can crunch data on a platform and in a device-agnostic manner. Ideally, development of meaningful AI will demand that machines obtain some form of human consciousness to create useful and powerful assistants.

While there has been rapid progress in hardware, storage and parallel-processing architectures, the field of artificial consciousness remains in its infancy stage. And much like the human body, this system is expected to carry out its functions and adapt to its user's requirements without the need of the user to go into minute details of its functioning. **EFY**